Assignment Day 4
C++ Programming

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Batch : February 2026

22. Declare function and call it by reference for swapping numbers

Program:

```
#include<iostream>
using namespace std;

// function to swap numbers using call by reference
void swapNum(int &num1, int &num2){
   int temp;
   temp = num1;
   num1 = num2;
   num2 = temp;
}

int main (){
   int num1, num2;
   cout << "Enter 2 numbers : ";
   cin >> num1 >> num2;

   cout << "Before swapping : " << num1 << "  " << num2 << endl;
   // function call by reference
   swapNum(num1 , num2);
   cout << "After swapping : " << num1 << "  " << num2 << endl;
   return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day4 % cd "/Users/a
ktop/cpp /day4/"swappingNumbers
Enter 2 numbers : 20 65
Before swapping : 20  65
After swapping : 65  20
(base) abhinyaay@Abhinays-MacBook-Air day4 %
```

23. Create function for printing an array and call it from main (Student array)

Program:

```
#include<iostream>
using namespace std;

void printArray(string student[] , int size) {
   cout << "Student array : ";
   for (int i = 0 ; i < size ; i++){
       cout << student[i] << "  ";
   }
}


int main (){

   int size;
   cout<< "Enter the size of student array  : ";
   cin >> size;

   string student[size];

   cout << "Enter " << size << "  student names : ";
   for (int i = 0 ; i<size; i++){
       cin >> student[i];
   }

   // function call
   printArray(student, size);

   return 0;
}

// Pawan Varun Prashant Ankit Abhishek
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day4 % cd "/Users/abhinyaa
day4/"printArray
Enter the size of student array  : 5
Enter 5  student names : Pawan Varun Prashant Ankit Abhishek
Student array : Pawan  Varun  Prashant  Ankit  Abhishek  %
(base) abhinyaay@Abhinays-MacBook-Air day4 %
```

24. Function for print factorial series with recursive

Program:

```
#include<iostream>
using namespace std;

// funtion to print factorila using recursion
int factorial (int n){
   // base condition
   if( n==0 || n==1){
       return 1;
   }
   // recursive call
   else {
       return n * factorial(n-1);
   }
}

int main (){
   int n;
   cout << "Enter a numer to find the factorial : ";
   cin >> n;

   // function calling and printing the value
   cout << "Factorial of " << n << " is :  " <<factorial(n);
   return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day4 % cd "/Users/a
y4/"factorial
Enter a numer to find the factorial : 5
Factorial of 5 is :  120%
(base) abhinyaay@Abhinays-MacBook-Air day4 %
```

25. Reverse string (char array) with recursion

Program:

```
#include<iostream>
#include<cstring>
using namespace std;

// function to reverse string by recursion

void reverse(char str[], int start, int end){
```

```
    // if start becomes greater than end and crosses the number of char
    // no character needs to be swapped
    if (start >= end)
        return;

    // swapping
    char temp;
    temp = str[start];
    str[start] = str[end];
    str[end] = temp;

    //recurive calling
    reverse(str, start + 1, end - 1);
}

int main (){
    char str[100];

    cout << "Enter a String : ";
    cin >> str;

    int n = strlen(str);

    cout << " String Before reversing :  " << str;

    // funtion call
    reverse(str, 0 , n-1);

    cout << "String After reversing  :  " << str;

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day4 % cd "/Users/a
/cpp /day4/"reverseString
Enter a String : moon
 String Before reversing :  moonString After reversing  :
(base) abhinyaay@Abhinays-MacBook-Air day4 % cd "/Users/a
/cpp /day4/"reverseString
Enter a String : pool
 String Before reversing :  poolString After reversing  :
(base) abhinyaay@Abhinays-MacBook-Air day4 %
```

26. Create Student class and add rollNo and name in it. Create Accept() func-tion which will take an array of Students as input. Write one more func-tion to display student data. This Display() function should receive Stu-dent array.

Program:

```
#include<iostream>
using namespace std;

// Student class function
class Student
{
   public:
       int rollNum;
       string name;
       int age;
       int marks;

   // function to accept data

   void accept ()
   {
       cout << "Roll Number : ";
       cin >> rollNum;

       cout << "Name : ";
       cin >> name;

       cout << "Age : ";
       cin >> age;

       cout << "Marks : ";
       cin >> marks;
   }

   // function to display data
   void display ()
   {
       cout << "Roll Number :  " << rollNum << endl;
       cout << "Name :  " << name << endl;
       cout << "Age :  " << age << endl;
       cout << "Marks :  " << marks << endl;
   }

};


int main (){
```

```
    int n;
    cout << "Enter the number of student : ";
    cin >> n;

    Student* s = new Student[n];

    for(int i = 0; i < n; i++)
    {
        cout << "Enter details of student " << i+1 << endl;
        s[i].accept();
    }

    cout << "Student Details"<<endl;

    for(int i = 0; i < n; i++)
    {
        cout << "Student " << i+1 << endl;
        s[i].display();
    }



    delete[]s; // freeing memory
    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day4 % cd "/Users/a
pp /day4/"studentclass
Enter the number of student : 4
Enter details of student 1
Roll Number : 101
Name : Ankit
Age : 23
Marks : 67
Enter details of student 2
Roll Number : 102
Name : Pawan
Age : 22
Marks : 98
Enter details of student 3
Roll Number : 103
Name : Varun
Age : 24
Marks : 88
Enter details of student 4
Roll Number : 104
Name : alok
Age : 22
Marks : 78
```

```
Student Details
Student 1
Roll Number :  101
Name :  Ankit
Age :  23
Marks :  67
Student 2
Roll Number :  102
Name :  Pawan
Age :  22
Marks :  98
Student 3
Roll Number :  103
Name :  Varun
Age :  24
Marks :  88
Student 4
Roll Number :  104
Name :  alok
Age :  22
Marks :  78
(base) abhinyaay@Abhinays-MacBook-Air day4 %
```

1. Accept n numbers from users and store them in to dynamically allocated array. Write a function to sort and display sorted array.

Program:

```
#include<iostream>
using namespace std;

// function to sort array
void sortArray(int *arr, int size){
   int temp;
   for (int i = 0 ; i < size ; i++){
       for (int j=i+1 ; j < size ; j++ ){
           if (arr[i] > arr[j])
           {
               temp = arr[i];
               arr[i] = arr[j];
               arr[j] = temp;
           }
       }
   }
}

// function to display array
void dispayArray(int *arr, int size){
   for (int i = 0 ; i < size ; i++){
```

```
        cout << arr[i] << "  ";
    }
}

int main (){
    int n;
    cout << "Enter number of elements: ";
    cin >> n;

    int *arr = new int[n];

    cout << "Enter numbers: "<<endl;
    for (int i = 0; i < n ; i++){
        cin >> arr[i];
    }

    // function call
    dispayArray(arr,n);
    cout<<endl;
    sortArray(arr,n);
    dispayArray(arr,n);

    delete[]arr;

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day4 % cd "/Users/abhinyaa
y4/"sortarray
Enter number of elements: 5
Enter numbers:
2 2874 1824 1 32875 8
2  2874  1824  1  32875
1  2  1824  2874  32875  %
(base) abhinyaay@Abhinays-MacBook-Air day4 %
```

2. Accept n characters from user and print is as string.

Program:

```
#include<iostream>
using namespace std;

int main(){
    int n;
    cout << "Enter number of characters : ";
```

```
    cin >> n;

    // dynamic memory allocation
    // n+1 for \0 stores in last at the string
    char *str = new char [n+1];

    cout << "Enter characters : ";
    for (int i = 0; i < n; i++){
        cin >> str[i];
    }

    str[n] = '\0';

    cout << "String is : " << str << endl;

    // free memory
    delete[] str;
    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day4 % cd "/Users/abhinyaa
pp /day4/"charToString
Enter number of characters : 5
Enter characters : h e l l o
String is : hello
(base) abhinyaay@Abhinays-MacBook-Air day4 %
```

3. Accept n numbers from user and write a function to find out average and display average.

Program:

```
#include<iostream>
using namespace std;

// function to find average
float findAvg(int arr[], int n){
    int sum = 0;
    for (int i = 0; i < n ; i++){
        sum = sum + arr[i];
    }

    return float(sum) / n;
}
```

```
int main(){
   int n;
   cout << "Enter number of elements : ";
   cin >> n;

   int arr[n];
   cout << "Enter number : ";
   for (int i = 0 ; i < n ; i++){
       cin >> arr[i];
   }

   float avg = findAvg(arr,n);

   cout << "Average  = " << avg << endl;
   return 0 ;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day4 % cd "/Users/abhinyaa
average
Enter number of elements : 5
Enter number : 10 20 30 40 50
Average  = 30
(base) abhinyaay@Abhinays-MacBook-Air day4 %
```

4. Accept n numbers from user and write a function to multiply each number by 2 and display result.

Program:

```
#include<iostream>
using namespace std;

// function to multiply each number by 2

void mulbytwo(int arr[], int size){
   for (int i = 0; i<size ; i++){
       arr[i] = arr[i]*2;
   }
}

// function to display

void displayArr(int arr[], int size){
   for (int i = 0 ; i <size ; i++){
       cout << arr[i] <<"  ";
```

```
    }
}

int main(){
    int n;
    cout << "Enter size of array : ";
    cin >> n;

    int arr[n];

    cout << "Enter number: ";
    for (int i=0; i < n ; i++){
        cin >> arr[i];
    }
    displayArr(arr,n);
    mulbytwo(arr,n);
    cout << endl;
    displayArr(arr,n);


    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day4 % cd "/Users/abhinyaa
lby2
Enter size of array : 5
Enter number: 10 20 30 40 50
10  20  30  40  50
20  40  60  80  100  %
(base) abhinyaay@Abhinays-MacBook-Air day4 %
```

27. Create Student class and add rollNo and name in it. Create Accept() func-tion which will take an array of Students as input. Write one more func-tion to display student data. This Display() function should receive Stu-dent array.

Program:

```
#include<iostream>
using namespace std;

class Student
{
    public:
    int rollNum;
    string name;
};
```

```
// funtion to accept student data
void Accept(Student s[], int n){
   for (int i = 0; i < n ; i++){
       cout << "Enter details : "<<endl;

       cout <<"Roll number : ";
       cin >> s[i].rollNum;
       cout <<"Name : ";
       cin >> s[i].name;
   }
}

// function to display student data
void Display(Student s[], int n ){
   cout << "Student Details : "<< endl;

   for (int i = 0; i < n; i++){
       cout << "Student "<< i+1 << endl;
       cout << "Roll Number : " << s[i].rollNum << endl;
       cout << "Name : " << s[i].name << endl;
   }
}

int main(){
   int n;
   cout <<" Enter number of students : ";
   cin >> n;

   Student s[n];

   Accept(s,n);
   Display(s,n);

   return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day4 % cd "/Users/abhinyaa
ktop/cpp /day4/"studentnameroll
 Enter number of students : 3
Enter details :
Roll number : 101
Name : Ankit
Enter details :
Roll number : 102
Name : Varun
Enter details :
Roll number : 103
Name : Prashant
Student Details :
Student 1
Roll Number : 101
Name : Ankit
Student 2
Roll Number : 102
Name : Varun
Student 3
Roll Number : 103
Name : Prashant
(base) abhinyaay@Abhinays-MacBook-Air day4 %
```